Materials Design Analysis Reporting (MDAR) Checklist for Authors

The MDAR framework establishes a minimum set of requirements in transparent reporting applicable to studies in the life sciences (see Statement of Task: doi:10.31222/osf.oio/9sm4x.). The MDAR checklist is a tool for authors, editors and others seeking to adopt the MDAR framework for transparent reporting in manuscripts and other outputs. Please refer to the MDAR Elaboration Document for additional context for the MDAR framework.

Materials

Antibodies	Yes (indicate where provided: page no/section/legend)	n/a
For commercial reagents, provide supplier name, catalogue number and RRID, if available	No antibodes detected. Please add identifiers for all resources where possible	

Cell Materials	Yes (indicate where provided: page no/section/legend)	n/a
Cell lines: Provide species information, strain. Provide accession number in repository OR supplier name, catalog number, clone number, OR RRID	Yes, 1 cell line detected, 0 RRID provided : E14TG2a Please add identifiers for all resources where possible	
Primary cultures: Provide species, strain, sex of origin, genetic modification status.	Not currently checked by SciScore	

Experimental Animals	Yes (indicate where provided: page no/section/legend)	n/a
Laboratory animals: Provide species, strain, sex, age, genetic modification status. Provide accession number in repository OR supplier name, catalog number, clone number, OR RRID	No organisms detected Please add identifiers for all resources where possible	
Animal observed in or captured from the field: Provide species, sex and age where possible	Not currently checked by SciScore	
Model organisms: Provide Accession number in repository (where relevant) OR RRID	See laboratory animals section for information.	

Plants and microbes	Yes (indicate where provided: page no/section/legend)	n/a
Plants: provide species and strain, unique accession number if available, and source (including location for collected wild specimens)	Not currently checked by SciScore	
Microbes: provide species and strain, unique accession number if available, and source	Not currently checked by SciScore	

Human research participants	Yes (indicate where provided: page no/section/legend)	n/a
Identify authority granting ethics approval (IRB or equivalent committee(s), provide reference number for approval.	Not detected.	
Provide statement confirming informed consent obtained from study participants.	Not detected.	
Report on age and sex for all study participants.	**Age:**not required. **Sex:**not detected.	

Design

Study protocol	Yes (indicate where provided: page no/section/legend)	n/a
For clinical trials, provide the trial registration number OR cite DOI in manuscript.	Not detected.	

Laboratory protocol	Yes (indicate where provided: page no/section/legend)	n/a
Provide DOI or other citation details if detailed step-by-step protocols are available.	Not detected.	

Experimental study design (statistics details)	Yes (indicate where provided: page no/section/legend)	n/a
State whether and how the following have been done, or if they were not carried out		
Sample size determination	not detected.	
Randomization	Sixteen clones of each sample were randomly selected for Sanger sequencing.	
Blinding	not detected.	
inclusion/exclusion criteria	not required.	

Sample definition and in-laboratory replication	Yes (indicate where provided: page no/section/legend)	n/a
State number of times the experiment was replicated in laboratory	Not detected.	
Define whether data describe technical or biological replicates	For each biological replicate fraction of CpG sites with at least one read was computed and only CGIs where all biological replicates had identified CpG sites were selected.	

Ethics	Yes (indicate where provided: page no/section/legend)	n/a
Studies involving human participants: State details of authority granting ethics approval (IRB or equivalent committee(s), provide reference number for approval.	Not detected.	
Studies involving experimental animals: State details of authority granting ethics approval (IRB or equivalent committee(s), provide reference number for approval.	Not detected.	
Studies involving specimen and field samples: State if relevant permits obtained, provide details of authority approving study; if none were required, explain why.	Not detected.	

Dual Use Research of Concern (DURC)	Yes (indicate where provided: page no/section/legend)	n/a
If study is subject to dual use research of concern, state the authority granting approval and reference number for the regulatory approval	Not currently checked by SciScore	

Analysis

Attrition	Yes (indicate where provided: page no/section/legend)	n/a
State if sample or data point from the analysis is excluded, and whether the criteria for exclusion were determined and specified in advance.	not required.	

Statistics	Yes (indicate where provided: page no/section/legend)	n/a
Describe statistical tests used and justify choice of tests.	All protein-coding genes were used as a background set and only GO enrichments with Benjamini-Hochberg adjusted q-value less than 0.001 were selected.	

Data availability	Yes (indicate where provided: page no/section/legend)	n/a
State whether newly created datasets are available, including protocols for access or restriction on access.	Materials availability Plasmids generated in this study will be made available upon request.	
If data are publicly available, provide accession number in repository or DOI or URL.	STARMethods Key resources table REAGENT or RESOURCE SOURCE IDENTIFIER Chemicals, peptides, and recombinant proteins Gelatin Sigma-Aldrich CatG1890 Trypsin-EDTA Gibco Cat15400-054 Dulbeccos modified Eagles medium Gibco Cat11960-044 Embryonic stem-cell FBS Gibco Cat16141-079 Penicillin-Streptomycin Gibco Cat15140-122 Sodium pyruvate Gibco Cat11360-070 2-mercaptoethanol Gibco Cat21985-023 GlutaMAX Gibco Cat35050-061 MEM Non-essential amino acids Gibco Cat11140-050 Mouse leukemia inhibitory factor EMD Millipore CatESG1106 CHIR99021 Sigma-Aldrich CatSML1046 PD0325901 Sigma-Aldrich CatPZ0162 Opti-MEM Gibco Cat11058-021 TranscriptAid T7 High Yield Transcription Kit Thermo Fisher Scientific CatK0441 GeneArt Platinum Cas9 Nuclease Invitrogen CatB25640 Lipofectamine LTX Invitrogen Cat15335-100 Puromycin Sigma-Aldrich CatP8833 Phire Tissue Direct PCR Master Mix Thermo Fisher Scientific CatF170L S-adenosyl-L-homocysteine Sigma-Aldrich CatA9384 S-adenosyl-L-methionine Sigma-Aldrich CatA7007 AdoButen (S-Adenosyl-S-crotyl-L-homocysteine) Prof. Elmar Weinhold (Stecher et al., 2009) NA AdoButyn (S-Adenosyl-S-but-2-ynyl-L-homocysteine) Dalhoff et al., 2006 NA AdoEnYn (S-Adenosyl-S-pent-4-yn-2-enyl-L-homocysteine) Peters et al., 2010 NA Ado-6-azide (S-Adenosyl-S-(6-azidohex-2-ynyl)- L-homocysteine) Masevicius et al., 2016 NA Ado-6-amine (S-Adenosyl-S-(6-aminohex-2-ynyl)-L-homocysteine) Masevicius et al., 2016 NA Ado-6-ethyne (S-Adenosyl-S-(oct-2,6-diynyl)-L-homocysteine) Lukinavicius et al., 2013 NA Ado-13-biotin Osipenko et al., 2017 NA Poly(dI-dC) * Poly(dI-dC) Sigma-Aldrich CatP4929 Poly(dG-dC) * Poly(dG-dC) Sigma-Aldrich CatP9389 methyl- 3H-S-adenosyl-L-methionine PerkinElmer CatNET155001MC g-32PATP PerkinElmer CatNEG502Z001MC T4 PNK Thermo Fisher Scientific CatEK0032 Geneticin Sigma-Aldrich CatA1720 PMSF Sigma-Aldrich Cat78830 cOmplete Mini EDTA-free Protease Inhibitor Cocktail Roche Cat4693159001 SatI restriction enzyme Thermo Fisher Scientific CatER1641 FastDigest BspTI Thermo Fisher Scientific CatFD0834 FastDigest NotI Thermo Fisher Scientific CatFD0596 FastDigest EcoRI Thermo	

Data availability	Yes (indicate where provided: page no/section/legend)	n/a
	Fisher Scientific CatFD0274 FastDigest SacI Thermo Fisher Scientific CatFD1134 FastDigest HhaI Thermo Fisher Scientific CatFD1854 FastDigest XbaI Thermo Fisher Scientific CatFD0685 Scintillation cocktail Rotiszint(r) Eco plus Carl Roth Cat0016.3 Proteinase K, recombinant, PCR grade Thermo Fisher Scientific CatEO0491 RNase A Thermo Fisher Scientific CatEN0531 Genomic DNA Clean & Concentrator-10 Kit Zymo Research CatD4011 Nuclease P1 from Penicillium citrinum Sigma-Aldrich CatN8630 FastAP Thermosensitive Alkaline Phosphatase Thermo Fisher Scientific CatEF0654 dCTP Thermo Fisher Scientific CatR0151 dGTP Thermo Fisher Scientific CatR0161 dATP Thermo Fisher Scientific Cat R0141 dTTP Thermo Fisher Scientific CatR0171 5-Methyl-dCTP Thermo Fisher Scientific CatR0431 Alkyne MegaStokes dye 608 Sigma-Aldrich Cat79249 Igepal CA-630 Sigma-Aldrich CatI8896 BCN-amine Sigma-Aldrich Cat745073 MTT Sigma-Aldrich CatM5655 Fast DNA End Repair Kit Thermo Fisher Scientific CatK0771 Klenow Fragment, exo- Thermo Fisher Scientific CatEP0422 T4 DNA Ligase Thermo Fisher Scientific CatEL0011 CuBr, 99.999 Sigma-Aldrich Cat254185-10G DMSO Sigma-Aldrich Cat472301 THPTA Sigma-Aldrich Cat762342-500MG dNTP Mix (2 mM each) Thermo Fisher Scientific CatR0241 Pfu DNA polymerase (recombinant) Thermo Fisher Scientific CatEP0502 Platinum SuperFi PCR Master Mix Thermo Fisher Scientific Cat12358010 Dynabeads MyOne C1 Streptavidin magnetic beads Thermo Fisher Scientific Cat65002 GeneJET PCR purification kit Thermo Fisher Scientific CatK0702 GeneJet NGS Cleanup kit Thermo Fisher Scientific CatK0851 DNA Clean & Concentrator-5 Kit Zymo Research CatD4013 MagJET NGS Cleanup and Size Selection Kit Thermo Fisher Scientific Cat K2821 Agilent High Sensitivity DNA Kit Agilent Cat5067-4626 Ion PI(tm) Hi-Q(tm) OT2 200 Kit Thermo Fisher Scientific Cat A26434 Ion PI(tm) Hi-Q(tm) Sequencing 200 Thermo Fisher Scientific Cat A26772 Ion PI(tm) Chip Kit v3 Thermo Fisher Scientific Cat A26771 GeneJet RNA purification kit Thermo Fisher Scientific CatK0731 dsDNase Thermo Fisher Scientific CatEN0771 RevertAid(tm) RT reverse transcriptase Thermo Fisher Scientific CatEP0441 2x SYBR(tm) Green PCR Master Mix Thermo Fisher Scientific CatK253 Phusion U Hot Start polymerase Thermo Fisher Scientific CatF555S EZ DNA Methylation-Gold Kit Zymo Research CatD5005 Deposited data Dnmt-TOP-seq data This work GEO: GSE182445 ChIP-seq data for Dnmt3a1 and Dnmt3b Weinberg et al., 2019 GEO: GSE118785 Raw gel images This work Mendeley Data: 10.17632kddxk2d7pc.1 Experimental models: Cell lines Mouse embryonic stem cells, E14TG2a ATCC CRL-1821 Mouse embryonic stem cells, E14TG2a Dnmt1 N1580A This work NA Mouse embryonic stem cells, E14TG2a Dnmt1-KO This work NA P. pastoris GS115 Thermo Fisher Scientific CatC18100 Oligonucleotides See Table S2 for oligonucleotide sequences Metabion NA Biotinylated alkyne-containing DNA oligonucleotide, 5-TXTTTTGTGTGGTTTGGAGACT GACTACCAGATGT AACA-Biotin X=C8-Alkyne-dT BaseClick NA Complementary priming strand with custom LNA modifications and phosphorothioate linkages at the 3 end, 5-	

Data availability	Yes (indicate where provided: page no/section/legend)	n/a
	TGTTACATCTGGTAGTCAGTCT CCAAACCACACAA Exiqon NA Recombinant DNA pSpCas9(BB)-2A-Puro (PX459) plasmid Addgene (Rohde et al., 2010) Cat62988 pDL2-14 plasmid Gerasimaite et al., 2009 NA pUC19 plasmid Thermo Fisher Scientific CatSD0061 pPIC3.5K Thermo Fisher Scientific CatV17320 pPIC3.5K-Dnmt1-dN Vilkaitis et al., 2005 NA Software and algorithms Cutadapt (3.4) Martin, 2011 https:cutadapt.readthedocs.io R (3.5) R project https:www.r-project.org FASTX (0.0.13) Hannon Lab, CSHL http:hannonlab.cshl.edufastx_toolkitindex.html BWA (0.7.17) Li and Durbin, 2009 http:bio-bwa.sourceforge.net PyMOL (1.7) PyMOL Molecular Graphics System by Schrodinger https:pymol.org liftOver UCSC genome browser store https:genome-store.ucsc.edu clusterProfile (3.10.1) Yu et al., 2012 http:bioconductor.orgpackagesreleasebiochtmlclusterProfiler.html Other Repeat Masker annotation Karolchik et al., 2004 https:genome.ucsc.edu CpG island annotation Karolchik et al., 2004 https:genome.ucsc.edu Gene annotation Frankish et al., 2019 https:www.gencodegenes.org Detailed bench protocol This work Methods S1 Resource availability Lead contact Further information and reasonable requests for resources and reagents should be directed to and will be fulfilled by the Lead Contact, Saulius Klimasauskas (saulius.klimasauskasbti.vu.lt).	
If publicly available data are reused, provide accession number in repository or DOI or URL, where possible.	Not detected.	

Code availability	Yes (indicate where provided: page no/section/legend)	n/a
For all newly generated code and software essential for replicating the main findings of the study:		
State whether the code or software is available.	Not detected.	
If code is publicly available, provide accession number in repository, or DOI or URL.	Not detected.	

Analysis

Adherence to community standards	Yes (indicate where provided: page no/section/legend)	n/a
MDAR framework recommends adoption of discipline-specific guidelines, established and endorsed through community initiatives. Journals have their own policy about requiring specific guidelines and recommendations to complement MDAR.		
State if relevant guidelines (eg., ICMJE, MIBBI, ARRIVE) have been followed, and whether a checklist (eg., CONSORT, PRISMA, ARRIVE) is provided with the manuscript.	Not currently checked by SciScore	